Quantification: 701_z36c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 19 |
| interior lacuna count | 11 |
| frame-edge lacuna count | 8 |

Interior lacunae only: mean, median and sample SD (n = 11)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1504.36 | 897 | 1753.25 | px^2 |
| major_axis_length_px | lacuna shape | 65.8954 | 54.1218 | 44.4863 | px |
| minor_axis_length_px | lacuna shape | 32.4977 | 24.7285 | 15.3599 | px |
| aspect_ratio | lacuna shape | 2.0209 | 1.7172 | 0.7947 | unitless |
| eccentricity | lacuna shape | 0.8034 | 0.8129 | 0.1253 | unitless |
| solidity | lacuna shape | 0.7542 | 0.7412 | 0.1221 | unitless |
| roots_count | headline | 10.0909 | 9 | 4.5267 | count |
| ring_length_r30_px | headline | 452.818 | 399 | 174.489 | px |
| ring_length_r60_px | reported | 1060.45 | 917 | 389.391 | px |
| owned_length_px | ownership-dependent | 1931.93 | 1276.12 | 1456.71 | px |
| edge_count | ownership-dependent | 140.091 | 86 | 108.58 | count |
| mean_edge_length_px | ownership-dependent | 14.3631 | 13.9313 | 1.5558 | px |
| perimeter_px | normalised | 235.331 | 179.622 | 166.75 | px |
| ring_area_r30_px2 | normalised | 7434.18 | 6781 | 3418.26 | px^2 |
| ring_area_r60_px2 | normalised | 17465.8 | 16160 | 7147.61 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 0.994 | 1 | 0.0198 | unitless |
| ring_density_r30 | normalised | 0.0640281 | 0.0553758 | 0.0191072 | px^-1 |
| ring_density_r60 | normalised | 0.0629206 | 0.0574034 | 0.0151312 | px^-1 |
| roots_per_100px_perimeter | normalised | 5.146 | 3.7911 | 2.4252 | per 100 px |
| ring_attached_length_r30_px | network v2 | 373.182 | 331 | 166.606 | px |
| ring_attached_length_r60_px | network v2 | 733.182 | 593 | 374.894 | px |
| ring_length_w_r30_px | network v2 | 512.676 | 461.031 | 197.925 | px |
| ring_length_w_r60_px | network v2 | 1197.82 | 1035.59 | 441.756 | px |
| sholl_crossings_r10 | network v2 | 9.5455 | 10 | 3.5879 | count |
| sholl_crossings_r20 | network v2 | 11.4545 | 10 | 4.3672 | count |
| sholl_crossings_r30 | network v2 | 13.4545 | 14 | 4.3672 | count |
| loop_count | loops, report only | 12.3636 | 5 | 16.6929 | count |
| loop_edge_share | loops, report only | 0.269 | 0.2595 | 0.1732 | share |
| ring_width_mean_r30_px | width | 6.0698 | 6.2087 | 1.3318 | px |

Per field
---------

| measure | value | units |
|---|---|---|
| analysed_area_px2 | 996229 | px^2 |
| total_skeleton_length_px | 59591 | px |
| canalicular_length_density_per_px | 0.0598166 | px^-1 |
| junction_count | 1887 | count |
| junction_density_per_px2 | 0.00189414 | px^-2 |
| lacuna_count | 19 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.907e-05 | px^-2 |
| skeleton_component_count | 845 | count |
| mean_component_length_px | 70.5219 | px |
| median_component_length_px | 12 | px |
| field_length_density_w_per_px | 0.0673777 | px^-1 |
| field_density_without_flagged_per_px | 0.0649711 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 92 | px |
| unattached_component_count | 7 | count |
| canalicular_length_density_attached_per_px | 0.0598166 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0599089 | px^-1 |
| loop_count | 325 | count |
| loop_edge_share | 0.2852 | share |
| edge_count | 4610 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.2674 | px |
| width_median_px | 6 | px |
| width_p10_px | 2.8284 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 38285 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4432 | 103.113 | 57.3862 | 1.7968 | 0.8308 | 0.8752 | 0.7863 | 35.4294 | 265.904 | True |
| 2 | 5447 | 154.686 | 50.1635 | 3.0836 | 0.946 | 0.7894 | 0.5665 | 62.9728 | 466.817 | True |
| 3 | 7233 | 130.507 | 74.6116 | 1.7492 | 0.8205 | 0.8879 | 1.2582 | 34.0104 | 969.455 | True |
| 4 | 1625 | 83.1546 | 31.6031 | 2.6312 | 0.925 | 0.7835 | 0.114 | 172.29 | 1011.39 | True |
| 5 | 804 | 41.5213 | 33.4623 | 1.2408 | 0.592 | 0.6949 | -0.5169 | 176.293 | 670.124 | False |
| 6 | 437 | 31.4519 | 18.3155 | 1.7172 | 0.8129 | 0.8918 | 1.4887 | 285.545 | 929.915 | False |
| 7 | 843 | 61.4626 | 24.6873 | 2.4896 | 0.9158 | 0.7274 | 1.4958 | 308.81 | 200.344 | False |
| 8 | 1362 | 78.3413 | 37.8814 | 2.0681 | 0.8753 | 0.558 | 0.4269 | 332.607 | 242.698 | False |
| 9 | 1410 | 54.1218 | 34.645 | 1.5622 | 0.7683 | 0.921 | -1.5551 | 413.697 | 828.063 | False |
| 10 | 9556 | 142.266 | 91.5717 | 1.5536 | 0.7653 | 0.9091 | 1.4115 | 461.649 | 963.906 | True |
| 11 | 481 | 29.6358 | 22.2683 | 1.3309 | 0.6599 | 0.8698 | 1.5129 | 580.47 | 966.104 | False |
| 12 | 6510 | 186.09 | 58.4873 | 3.1817 | 0.9493 | 0.6768 | -1.566 | 684.663 | 733.562 | False |
| 13 | 2377 | 86.3084 | 62.7585 | 1.3752 | 0.6865 | 0.5865 | -0.3235 | 741.99 | 862.039 | False |
| 14 | 994 | 72.7028 | 20.0045 | 3.6343 | 0.9614 | 0.7412 | 1.4325 | 768.803 | 413.569 | False |
| 15 | 1166 | 44.1476 | 35.3039 | 1.2505 | 0.6004 | 0.9276 | 0.2819 | 791.344 | 1008.62 | True |
| 16 | 897 | 53.6979 | 24.7285 | 2.1715 | 0.8877 | 0.766 | -1.5247 | 884.094 | 479.08 | False |
| 17 | 433 | 29.5156 | 20.2362 | 1.4586 | 0.728 | 0.8625 | 0.4576 | 943.721 | 446.413 | False |
| 18 | 3880 | 120.945 | 42.3369 | 2.8567 | 0.9367 | 0.9282 | -1.4866 | 1004.7 | 384.256 | True |
| 19 | 2460 | 98.2976 | 35.4294 | 2.7745 | 0.9328 | 0.9004 | 1.5163 | 1009.82 | 777.268 | True |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 10 | 453 | 1162 | 2764.7 | 194 | 14.251 | 6.5032 |
| 2 | 14 | 683 | 1606 | 2690.02 | 184 | 14.6196 | 6.5137 |
| 3 | 11 | 429 | 835 | 1819.53 | 145 | 12.5485 | 6.7812 |
| 4 | 9 | 333 | 676 | 647.115 | 49 | 13.2064 | 6.1216 |
| 5 | 8 | 443 | 1213 | 1932.6 | 158 | 12.2317 | 5.8682 |
| 6 | 7 | 291 | 894 | 1276.12 | 86 | 14.8386 | 7.4536 |
| 7 | 6 | 225 | 495 | 289.522 | 16 | 18.0951 | 4.8623 |
| 8 | 10 | 399 | 905 | 734.512 | 53 | 13.8587 | 6.9731 |
| 9 | 6 | 365 | 917 | 816.541 | 52 | 15.7027 | 4.7742 |
| 10 | 9 | 454 | 892 | 911.524 | 71 | 12.8384 | 6.4637 |
| 11 | 4 | 363 | 825 | 1085.05 | 74 | 14.6628 | 6.2087 |
| 12 | 17 | 766 | 1721 | 2006.11 | 144 | 13.9313 | 4.6308 |
| 13 | 14 | 439 | 1150 | 4008.34 | 274 | 14.629 | 4.0253 |
| 14 | 17 | 714 | 1652 | 4055.69 | 306 | 13.2539 | 6.9086 |
| 15 | 6 | 284 | 736 | 846.968 | 61 | 13.8847 | 6.0351 |
| 16 | 13 | 627 | 1282 | 4131.21 | 310 | 13.3265 | 6.9924 |
| 17 | 9 | 349 | 611 | 915.553 | 68 | 13.464 | 8.0706 |
| 18 | 6 | 489 | 1002 | 5679.42 | 438 | 12.9667 | 7.2113 |
| 19 | 4 | 260 | 778 | 207.267 | 12 | 17.2723 | |

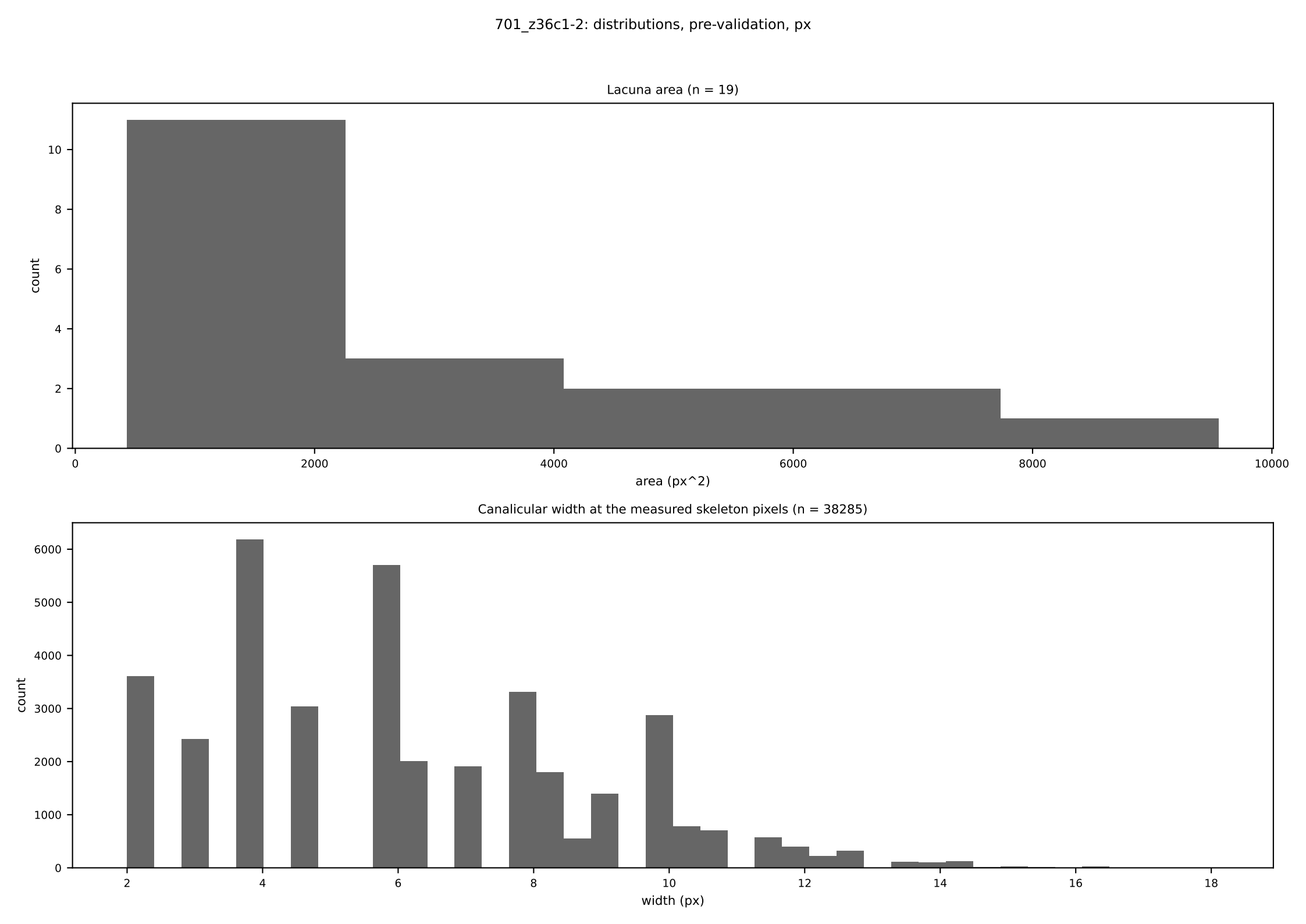
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 375.375 | 8196 | 18822 | 0.7168 | 0.6687 | 0.0552709 | 0.0617363 | 2.664 |
| 2 | 544.968 | 12171 | 26429 | 0.8379 | 0.7759 | 0.056117 | 0.0607666 | 2.569 |
| 3 | 492.416 | 6624 | 12045 | 0.4791 | 0.4258 | 0.0647645 | 0.0693234 | 2.2339 |
| 4 | 303.842 | 5339 | 10334 | 0.5936 | 0.5634 | 0.0623712 | 0.0654151 | 2.9621 |
| 5 | 227.635 | 6849 | 19116 | 1 | 1 | 0.064681 | 0.0634547 | 3.5144 |
| 6 | 94.468 | 5255 | 15574 | 1 | 1 | 0.0553758 | 0.0574034 | 7.4099 |
| 7 | 173.794 | 4559 | 10794 | 1 | 1 | 0.0493529 | 0.0458588 | 3.4524 |
| 8 | 267.019 | 7579 | 18429 | 1 | 1 | 0.0526455 | 0.0491074 | 3.745 |
| 9 | 158.267 | 6781 | 16160 | 1 | 1 | 0.0538269 | 0.056745 | 3.7911 |
| 10 | 551.263 | 10755 | 18353 | 0.7333 | 0.694 | 0.0422129 | 0.0486024 | 1.6326 |
| 11 | 101.225 | 5450 | 12434 | 1 | 0.9344 | 0.0666055 | 0.0663503 | 3.9516 |
| 12 | 628.95 | 16115 | 34097 | 1 | 1 | 0.0475334 | 0.0504737 | 2.7029 |
| 13 | 461.47 | 10803 | 23090 | 1 | 1 | 0.0406369 | 0.0498051 | 3.0338 |
| 14 | 198.309 | 7886 | 20602 | 1 | 1 | 0.0905402 | 0.0801864 | 8.5725 |
| 15 | 148.125 | 4251 | 11046 | 0.6223 | 0.5852 | 0.0668078 | 0.0666304 | 4.0506 |
| 16 | 179.622 | 6400 | 14669 | 1 | 1 | 0.0979687 | 0.0873952 | 7.2374 |
| 17 | 97.8823 | 4099 | 7159 | 1 | 1 | 0.0851427 | 0.0853471 | 9.1947 |
| 18 | 315.907 | 6243 | 12280 | 0.6169 | 0.5909 | 0.0783277 | 0.0815961 | 1.8993 |
| 19 | 302.007 | 5690 | 14075 | 0.5535 | 0.5397 | 0.0456942 | 0.0552753 | 1.3245 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 392 | 950 | 507.233 | 1321.4 | 12 | 12 | 11 |
| 2 | 597 | 1129 | 769.571 | 1831.75 | 16 | 14 | 18 |
| 3 | 421 | 700 | 487.019 | 945.11 | 12 | 8 | 9 |
| 4 | 320 | 493 | 371.664 | 758.955 | 9 | 7 | 7 |
| 5 | 372 | 884 | 508.617 | 1375.79 | 11 | 9 | 18 |
| 6 | 274 | 593 | 338.22 | 1003.15 | 7 | 9 | 9 |
| 7 | 128 | 391 | 250.965 | 560.458 | 3 | 9 | 6 |
| 8 | 331 | 424 | 461.031 | 1035.59 | 9 | 10 | 9 |
| 9 | 214 | 356 | 409.534 | 1031.16 | 10 | 7 | 14 |
| 10 | 388 | 544 | 513.304 | 993.08 | 11 | 14 | 11 |
| 11 | 302 | 606 | 403.149 | 923.382 | 9 | 10 | 14 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 12 | 574 | 1108 | 857.453 | 1935 | 16 | 22 | 18 |
| 13 | 319 | 478 | 483.576 | 1280.99 | 11 | 12 | 14 |
| 14 | 640 | 1501 | 821.955 | 1895.63 | 12 | 17 | 18 |
| 15 | 234 | 539 | 323.25 | 826.169 | 6 | 7 | 8 |
| 16 | 617 | 1148 | 708.931 | 1438.21 | 12 | 12 | 18 |
| 17 | 334 | 576 | 396.007 | 696.63 | 5 | 9 | 10 |
| 18 | 475 | 843 | 541.019 | 1118.6 | 12 | 14 | 10 |
| 19 | 176 | 405 | 282.966 | 878.838 | 7 | 10 | 13 |


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
2
4
6
8
10
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 38285)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16
18